ReliableML Method Note: Statistical Reason Audit

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Abstract

The reason audit combines labeled validity, perturbation-based feature reliance, subgroup and time stability, counterfactual sensitivity, logistic associations, and explanation alignment. It investigates potential shortcuts; it does not prove causality or correct reasons.

1 Audit design

Use labeled, representative holdout rows not used for fitting or model selection, plus training/reference background data. The report includes bootstrap intervals for predictive measures such as accuracy, multiclass Brier score, and top-label calibration error. This addresses whether predictions perform adequately on the evaluated sample, independently of what explanation methods say.

2 Feature reliance

For each feature, the audit perturbs held-out values using donor rows and measures the increase in held-out Brier loss. Conditional donor selection can use nearest neighbors over user-specified numeric covariates; this is a conditional-distribution approximation and can fail in sparse regions. Without conditional covariates, marginal perturbation may construct unrealistic combinations when features are correlated. Bootstrap intervals and repeated estimates quantify sampling variability but not all sources of uncertainty.

3 Stability and sensitivity

Optional groups and time periods report reliance across slices. Optional counterfactual replacements report prediction-flip rates and probability changes. The user is responsible for supplying domain-valid changes; arbitrary edits do not represent plausible interventions. Negative-control features and features unavailable at prediction time produce warnings, not definitive proof of leakage or shortcut learning.

4 Association and explanation alignment

For binary or selected one-vs-rest outcomes, an unregularized logistic model reports standardized-feature odds ratios, confidence intervals, p-values, and Benjamini–Hochberg adjusted p-values. These quantify conditional associations with observed labels. The audit can compare row-level

signed Shapley contributions with logistic log-odds contributions and compare supplied importance mappings with global Shapley ranks. Agreement is not a test of causality and can occur under confounding or shared bias.

5 Interpretation

No single score in the audit certifies correct-for-correct-reasons prediction. Conclusions depend on holdout representativeness, label quality, measurement, feature availability, dependence structure, calibration, and causal/domain knowledge. Current bootstrap procedures treat rows as independent; grouped or time-dependent data may need block or cluster resampling. Audits are computationally intensive because they repeatedly perturb data and evaluate the model.

6 Interface

```
audit = model.audit_reasons(
    X_holdout, y_holdout, positive_label=1,
    conditional_on=["age", "region"],
    groups={"sex": sex_holdout},
    time_periods=month_holdout,
    counterfactuals={"income": plausible_income_changes},
)
print(audit.predictive_validity)
print(audit.feature_reliance)
print(audit.logistic_association)
```

7 Further reading

References

[1] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 2017. https://arxiv.org/abs/1705.07874.

[2] M. Hardt, E. Price, and N. Srebro. Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems*, 2016. https://arxiv.org/abs/1610.02413.

[3] ReliableML reason audit assumptions and interfaces: https://github.com/Jabualani/ML-Reliabilty.